

Appendix: validations and work beyond the brief

Companion to report.pdf (3 pages). Every number here comes from files in the repo or the HF model repo (eval/, analysis/), regenerated by `python report/build_report.py`.

A. Work beyond the brief

Item	What it adds
Generalization analysis on 2 x 1,000 eval-only sentences	The 60-sentence dev slices cannot show the gap is real (CI includes zero); the larger samples, length and rare-word buckets and reference loss separate length, vocabulary and register (report section 5, section C below)
Bootstrap confidence intervals	Per dev slice and for the seen minus unseen gap
Checkpoint-set and decoding studies	3 checkpoint sets and an 11-point beam x length-penalty grid, extended when the optimum sat at the grid edge
Resumable training for free Colab	Atomic full-state checkpoints, deterministic batch order, wall-clock budget across sessions; survived two interruptions
Self-contained Hugging Face package	Weights (safetensors), tokenizer, inference code, model card with real metrics; loads with 3 dependencies
W&B report	Loss, LR, throughput, dev BLEU/chrF per slice, sample translations, config (scripts/wandb_report.py)
Colab notebook + setup guide	Secrets handling, Drive checkpoints, throughput check, analysis and Hub push cells (docs/COLAB_SETUP.md)
Reproducible reporting	report.pdf and this appendix are generated from result files by one script
Prepared ablations	RoPE vs sinusoidal and 6/6 configs ready to run (configs/)

B. Validations performed, with evidence

Check	What was done	Result
Model correctness (unit tests)	tests/test_sanity.py: (1) changing a later target token never changes earlier logits (causal mask); (2) appending padding to the source never changes the output, for RoPE and sinusoidal; (3) a tiny model memorises 8 random pairs, then greedy reproduces them, beam(k=1, alpha=0) equals greedy exactly, and beam(k=4) recovers them; (4) normalisation of curly quotes, guillemets, NBSP and ellipsis; (5) sentence splitter incl. abbreviations such as 'M.'	5/5 pass on Mac (CPU) and on the Colab T4 before training
Metric fidelity	src/metrics.py imports the official score.py functions instead of re-implementing them; its OVERALL was compared with running score.py on the same prediction file	Identical: 7.11 (smoke model) and 39.94 (final model)
Data pipeline reproducibility	prepare_data.py run independently on the Mac and on Colab from the same seed	Identical filter counts on both: 923,274 of 1,000,000 kept, every drop reason equal
Tokenizer reproducibility	Compared the SentencePiece model rebuilt on the Mac (8 threads) with the one trained on Colab (2 threads)	Not identical: 12 of 16,000 pieces differ, which shifts piece ids. Fixed by pinning 2 threads; the trained tokenizer ships with every checkpoint and the HF model
Train/eval contamination	Exact-match check of dev and test sources against the training sources, before and after normalisation	0 dev and 1 test source in raw train; 3 pairs after normalisation, all removed before training
Data-use audit (opus_books)	grep of every dataset load in src/: opus_books appears only in src/analyze.py, run after all model and decoding choices were fixed; dev/test sentences removed from the sample	Never used for training, tokenizer, checkpoint choice or decoding tuning
Resume correctness	Planned kill-and-resume test before the real run: smoke run stopped at step 600, resumed to 800	Caught a real bug (RNG state moved to GPU by map_location; fixed). After the fix: resumed at the exact batch, loss continued (5.82 -> 5.71 -> 5.60), LR schedule and hours carried over
Resume in production	Main run interrupted twice on Colab (disconnect at step ~2,200; GPU loss at step 9,000)	Both resumed from the last checkpoint into the same W&B run (5ks4crpz); no data repeated or skipped
Throughput-based model choice	300-step benchmark of the 6/6 model on the T4 against a threshold fixed before measuring (12k target tok/s)	11.5k tok/s, so switched to 6/3 (13.6k tok/s measured in the main run)
Training health	Validation loss (opus-100 validation, never used for selection) and dev scores every 2,000 steps	Val loss 4.11 -> 2.65 -> 2.28 -> 2.14 while dev OVERALL rose 14.5 -> 30.8 -> 35.4 -> 37.6: no overfitting; still improving
Checkpoint selection	Greedy dev OVERALL for step 9k alone vs averages of 7k-9k and 5k-9k	38.14 / 38.49 / 37.66: averaging the last 3 helps; including early checkpoints hurts. Chose 7k-9k
Decoding selection	Grid on dev: greedy; beam 4 and 5 x alpha 0.6/1.0/1.4; extended to alpha 1.8/2.2 because the best value sat at the grid edge (beam 8	Greedy 38.49 -> beam 5, alpha 1.8: 39.94 (peak; alpha 2.2 drops to 39.83). Gains

	not run: memory pressure, and 4 -> 5 gained < 0.4)	above alpha 1.4 are within noise
Statistical uncertainty	1,000-resample bootstrap 95% CIs per dev slice, and for the seen minus unseen_domain gap	Dev gap 7.9 BLEU, CI [-1.8, 17.5]: not significant on 60 sentences, so the gap was re-measured on 1,000 sentences per domain (31.6 vs 15.8 BLEU, greedy)
Submission file	test_predictions.json checked against sample_submission.json	All 330 ids present, none extra, 0 empty, 0 curly quotes
Long-input policy	Splitting only triggers above 128 subword tokens (never on dev); compared split vs no split on long eval-only sentences	23 such sentences: chrF 41.2 split vs 39.5 not split (BLEU 19.6 vs 19.0); kept as the default
Failure-mode claims	Each failure claim in the report checked against the 2,000 eval-only outputs before writing it: repetition counted (a 3-gram repeated 3+ times); literary past tenses inspected; rare-word buckets made disjoint after finding a boundary double count	Repetition 0.5% / 0.4% of outputs (in-domain / books); an over-strong 'tense flattened' claim and a wrong 'both domains fall with rare words' caption were corrected before publishing
Hugging Face model integrity	SHA-256 of model.safetensors and spm.model on the Hub compared with the local export; sizes for all other files	All 12 files match
Hugging Face model usability	The exported package loaded with only torch, sentencepiece and safetensors (no repo code) via fr_en_transformer.load_translator, as in the model card, and run on CPU	Loads and translates 3 sentences in 0.2 s, e.g. 'Ne t'inquiète pas !' -> 'Don't worry!'
Secrets and confidentiality	Every commit scanned for the HF and W&B key patterns; the brief (PDF) and plan are git-ignored; keys only in a local .env and Colab Secrets; a repo-only deploy key for GitHub	0 secrets in the public repo

C. Score by source length and by rare-word share (eval-only sets, n=1,000 each)

opus-100 test, by length (words)

bucket	n	BLEU	chrF
1-10	441	31.9	48.7
11-20	251	30.3	51.8
21-30	138	33.1	57.4
31-45	100	31.4	57.3
46+	70	30.6	55.6

opus_books sample, by length

bucket	n	BLEU	chrF
1-10	315	17.4	35.8
11-20	305	16.9	38.8
21-30	181	15.5	38.9
31-45	105	14.9	39.3
46+	94	15.3	39.0

opus-100 test, by share of rare source words

bucket	n	BLEU	chrF
none	749	31.9	51.6
up to 10%	145	29.9	53.0
over 10%	106	33.9	53.9

opus_books sample, by share of rare source words

bucket	n	BLEU	chrF
none	468	17.8	38.4
up to 10%	342	16.4	39.8
over 10%	190	9.3	33.4

D. Lowest- and highest-scoring dev sentences per slice (sentence chrF)

seen: lowest / highest sentence chrF

- **chrF 6** dev_00055
SRC: Date de modification : 2001-09-24 Haut de la page Avis importants
REF: His presentation provided an overview of the Canadian Shellfish Sanitation Program and its elements; aquaculture drug residue testing (focused on therapeutant use), the Healthy Salmon Program, and Pilot Compliance, Certification and Accreditation. <- Previous First Next ->
HYP: Date modified: 2001-09-24 Top of Page Important Notices
- **chrF 9** dev_00054
SRC: - Allez, file.
REF: - Run along, now.
HYP: - Come on, come on.
- **chrF 11** dev_00057
SRC: Ça avait tout d'une affaire non classée.
REF: This should have been an X File.
HYP: It was all a non-class case.
- **chrF 13** dev_00053
SRC: -Vous parlez comme ce Toussaint Louverture.
REF: Dondon, Gros-Morne and Plaisance. -Impossible! -It's true.
HYP: - You're talking like this Allowance.
- **chrF 100** dev_00033
SRC: Projet de résolution I Préparatifs du douzième Congrès des Nations Unies pour la prévention du crime et la justice pénale
REF: Draft resolution I Preparations for the Twelfth United Nations Congress on Crime Prevention and Criminal Justice
HYP: Draft resolution I preparations for the twelfth United Nations Congress on Crime Prevention and Criminal Justice
- **chrF 100** dev_00017
SRC: Tu pourras me conduire à l'aéroport demain ?

REF: Can you drive me to the airport tomorrow?
HYP: Can you drive me to the airport tomorrow?

long: lowest / highest sentence chrF

- **chrF 24** dev_00062
SRC: Le taux d'emploi à plein temps toute l'année augmente avec le niveau de scolarité et diminue avec l'âge, tandis que le taux officiel de chômage augmente avec l'âge et tombe avec le niveau d'éducation et le revenu du ménage.
REF: Further, the incidence of respondents who were unemployed and looking for work fell between the first post-program week and the time of the survey (37 to 29 percent, respectively).
HYP: The full-time employment rate increases with the level of education and decreases with age, while the official unemployment rate increases with age and falls with the level of education and household income.
- **chrF 30** dev_00067
SRC: Et chacun allait se faire inscrire dans sa ville d'origine. Joseph, lui aussi, quitta la ville de Nazareth en Galilée, pour monter en Judée, à la ville de David appelée Bethléem, car il était de la maison et de la descendance de David.
REF: This first census was taken while Quirinus was governor of Syria. Everyone had to be registered in his own town. So everyone set out for his own city; Joseph too set out from Nazareth of Galilee. As he belonged to the family of David, being a descendant of his, he went to Judea to David's town of Bethlehem to be registered with Mary, his wife, who was with child.
HYP: And each was going to be registered in his original city. Joseph, too, left the city of Nazareth in Galilee, to climb to Judah, to the city of David called Bethlehem, because he was house and the descendance of David.
- **chrF 41** dev_00068
SRC: Les musées les plus populaires de Paris peuvent être assez bondés pour l'occasion, il est donc préférable d'essayer d'y aller le matin ou en fin d'après-midi. Vous pouvez également décider de visiter des musées parisiens un peu moins connus (mais tout aussi fabuleux) comme le musée des Arts et Métiers ou le musée Rodin.
REF: You can also choose to visit one of the slightly less famous (but still absolutely fantastic) museums in Paris such as the Musée des Arts et Métiers or Musée Rodin.
HYP: Paris's most popular museums can be fairly bonded for the occasion, so you can also decide to visit Paris museums a little less known (but also fabulous) such as the Museum of Arts and Meters or the Rodin Museum.
- **chrF 42** dev_00074
SRC: Les députés sont favorables à la réduction à zéro de l'intervention pour les céréales (sauf pour le blé), et plus généralement, à ce que l'actuel système d'intervention soit remplacé par un filet de sécurité dans les situations de crise qui risquent de se multiplier notamment du fait du changement climatique.
REF: MEPs are in favour of lowering the intervention thresholds for market crops to '0' (except for wheat) and, more broadly, replacing the current system by a safety net for crises, which are likely to be more frequent owing to global warming.
HYP: Members are in favour of reducing the response to cereals (except for wheat), and more generally, to ensure that the current intervention system is replaced by a security net in crisis situations which may increase, in particular, due to climate change.
- **chrF 91** dev_00079
SRC: Amendement déposé par Marcello Vernola, Amalia Sartori, Guido Podestà, Giuseppe Castiglione, Renato Brunetta, Lorenzo Cesa, Gianni De Michelis, Miroslav Mikolášik + Alessandro Foglietta, Adriana Poli Bortone, Cristiana Muscardini, Sergio Berlato Amendement 533 Article 12, paragraphe 4, alinéa 1 4.
REF: Amendment by Marcello Vernola, Amalia Sartori, Guido Podestà, Giuseppe Castiglione, Renato Brunetta, Lorenzo Cesa, Gianni De Michelis, Miroslav Mikolášik, Alessandro Foglietta, Adriana Poli Bortone, Cristiana Muscardini and Sergio Berlato Amendment 533 Article 12, paragraph 4, subparagraph 1 4.
HYP: Amendment by Marcello Vernola, Amalia Sartori, Guido Podestà, Giuseppe Castiglione, Renato Brunetta, Lorenzo Cesa, Gianni De Michelis, Miroslav Mikolášik + Alessandro Foglitta, Adrana Pol Bortone, Cristiana Muscardini, Sergio Berlato Amendment 533 Article 12, paragraph 4, subparagraph 1, 4.
- **chrF 96** dev_00083
SRC: The Office also received allegations regarding the mass detention of 29 people in August by members of the Attorney-General's Office and the DAS in Argelia, Marinilla, Santuario and Medellín (Antioquia).
REF: The Office also received allegations regarding the mass detention of 29 people in August by members of the Attorney-General's Office and the DAS in Argelia, Marinilla, Santuario and Medellín (Antioquia).
HYP: The Office also received allegations regarding the mass detention of 29 people in August by members of the Attorney-General's Office and the DAS in Argentina, Marinilla, Santuario and Medellín (Anioquia).

unseen_domain: lowest / highest sentence chrF

- **chrF 12** dev_00125
SRC: CHAPITRE VI ON FRAPPE AU CARREAU
REF: VI SOME ONE KNOCKS AT THE WINDOW
HYP: CHAPTER VI ON THE CARREAU
- **chrF 14** dev_00139
SRC: C'est ainsi, du moins, que j'imagine aujourd'hui notre arrivée.
REF: Thus to-day I picture our arrival.
HYP: That is, at least, that I guess today.
- **chrF 16** dev_00117
SRC: Un coup brusque au carreau nous fit lever la tête.
REF: A sharp knock on the window made us look up.
HYP: A sudden shot at the car was lifting our head.
- **chrF 17** dev_00104
SRC: Ils nous avaient aperçus et ils nous huaient de la belle façon.
REF: They had spotted us and hooted like anything.
HYP: They've seen us, and they're in love with us.
- **chrF 80** dev_00134
SRC: Ne t'inquiète pas !
REF: Don't worry.'
HYP: Don't worry!
- **chrF 89** dev_00116
SRC: Le quatrième jour fut un des plus froids de cet hiver-là.

REF: The fourth day was one of the coldest of that winter.
HYP: The fourth day was one of the coldest of this winter.